TIZANIDINE

Hypotension

Do not stop suddenly

CHEMICAL NAME 5-chloro-N-(4,5-dihydro-1H-imidazol-2-yl)-2,1,3-benzothiadiazol-4-amine (as hydrochloride)

CLASS Centrally acting α2-adrenergic agonist muscle relaxant

TRADE NAMES Zanaflex · Sirdalud · Tizan

THERAPEUTIC USES / INDICATIONS

- Spasticity due to multiple sclerosis or spinal cord injury.
- Painful muscle spasm (some countries).
- Tension-type headache (off-label).

ADMINISTRATION

- **PO** tablets or capsules.
- Take consistently with or without food (food changes absorption).
- Titrate gradually; taper when stopping.

DOSAGES

- **Spasticity:** start 2 mg at night; increase by 2 mg every 3–4 days.
- Usual 12–24 mg/day in 3–4 doses; max 36 mg/day.
- **Muscle spasm:** 2–4 mg three times daily.
- Renal impairment (CrCl <25): start 2 mg once daily.

PHARMACOKINETICS

- **Onset:** 1–2 h. **Peak:** 1–2 h. **Duration:** 3–6 h.
- Extensive first-pass metabolism by CYP1A2.
- Excreted in urine. **t½:** 2.5 h.

PHARMACODYNAMICS

- **α2-adrenergic agonist** in spinal cord → ↓ release of excitatory neurotransmitters.
- Reduces spasticity and muscle tone; also lowers BP and causes sedation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, fatigue, dizziness.
- Dry mouth.
- Muscle weakness.

ADVERSE EFFECTS (SERIOUS)

- Hypotension and bradycardia.
- Hepatotoxicity (raised liver enzymes, hepatitis).
- Hallucinations.
- Rebound hypertension and tachycardia on abrupt withdrawal.

CONTRAINDICATIONS & CAUTIONS

- Use with fluvoxamine or ciprofloxacin (strong CYP1A2 inhibitors).
- Significant hepatic impairment.
- **Caution:** renal impairment, elderly, hypotension, cardiac disease.

KEY INTERACTIONS

- **Fluvoxamine, ciprofloxacin:** contraindicated (↑ tizanidine 10-fold).
- **Antihypertensives:** additive hypotension.
- **Oral contraceptives:** ↑ tizanidine levels.
- **Alcohol, sedatives:** ↑ sedation.

NURSING MANAGEMENT

- Monitor BP and HR, especially after dose changes.
- Check LFTs at baseline and at 1, 3 and 6 months.
- Assess muscle tone and spasticity response.
- Fall precautions; rise slowly.
- Taper dose when stopping.

PATIENT EDUCATION

- Rise slowly from lying or sitting.
- Do not stop suddenly.
- Avoid alcohol; may cause drowsiness.
- Report yellow skin, dark urine or fainting.

KEY POINTS

- α2 agonist (like clonidine) for spasticity.
- Avoid with ciprofloxacin/fluvoxamine.
- Watch for hypotension and liver toxicity.
- Taper to stop — rebound hypertension.